SkyPass: A Machine Learning and Exact Optimisation Framework for Weather-Aware Satellite Pass Scheduling at Small Ground Stations

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Abstract

Observation planning at a small ground station is split across unconnected tools that predict passes, judge visibility and leave cloud and conflicts to the operator. This work examines whether an integrated planner with learned components increases the number of passes observed under clear sky. SkyPass carries orbital elements through SGP4 propagation, pass extraction, visibility scoring, cloud-forecast calibration against reanalysis and exact conflict resolution by a dynamic program linear in the candidates and the nightly limit. Two learned components replace hand-built approximations: segmentation of whole-sky camera images as a sub-hourly cloud nowcast, and gradient-boosted trees that predict apparent magnitude from pass geometry, trained on 17 216 measured magnitudes. Pass extraction agrees with an independent implementation to 0.055 s using 23.7× fewer propagator evaluations, and the scheduler matched exhaustive search on all 2000 instances. Over 60 days, a station required to observe every night gained nothing from the forecast, whereas one free to choose its nights raised realised yield by 19.0% (expected value) and 37.6% (clear-sky count), and calibration raised the gain in every regime. Learned segmentation estimated each image's cloud fraction to within 1.2 percentage points, against 3.5 for the best colour-ratio rule, and the learned photometry cut the magnitude error for unseen satellites from 1.51 to 0.83 mag but did not transfer to a later period with new designs. Forecast value depends on the decision it informs; the learned components help where their training data match deployment.

Keywords: satellite pass prediction, ground station scheduling, interval scheduling, forecast calibration, cloud segmentation, machine learning

1 Introduction

Small ground stations, such as university laboratories, amateur-radio clubs and individual optical observers, face a recurring planning problem. Several hundred satellites are potentially observable and generate thousands of passes in a week, of which only a few dozen merit the operator's time, and the station attends to one object at a time.

Existing tools address only parts of this. Pass predictors compute rise and set times from two-line elements (TLEs) and the SGP4 propagator [1, 2], some with an illumination test; a few show a weather indication beside a pass, but to our knowledge none let the forecast *rank* passes or resolve overlapping passes.

Weather is the more consequential omission. At the site used as a running example (Chennai, India, 12.92° N), mean observed cloud over optically viable passes was 88%, under which most geometrically favourable passes are not observable, yet a predictor ranking by culmination elevation recommends an overcast night. Once several satellites are tracked their passes also overlap and a single-aperture station has to choose; the literature treats that from the operator side, as spacecraft tasking and range scheduling [3–5], with constraints a small station does not have.

This paper shows that the small-station problem admits a substantially simpler exact formulation, and that the treatment of weather is what improves realised outcomes. The contributions are: an integrated pipeline accounting for the passes removed by each physical constraint; a validated pass extractor; exact capacity-constrained conflict resolution requiring neither a solver nor a metaheuristic; a weather claim verified against ERA5 reanalysis [6]; two negative results, namely that weather-awareness yields no benefit under a fixed nightly quota and that the uncalibrated forecast degrades performance there; and two learned components, a cloud nowcast from whole-sky images and a data-driven photometric model, each evaluated against classical baselines on identical data.

2 Related Work

SGP4, defined in Spacetrack Report No. 3 [7] and maintained in a revised reference implementation [8], remains with its two-line element sets the standard basis for rapid position prediction [1, 2]. Accuracy is limited by the elements rather than the propagator: errors grow with element-set age, as measured against on-board GPS [9]. Whether a satellite is *seen* depends on illumination and photometry, treated with a conical Earth-shadow model [1] and a diffuse-sphere phase function [10]; the brightness of large constellations has meanwhile become a concern for ground-based astronomy in its own right [11]. Measured satellite magnitudes have recently become available in bulk through the IAU CPS SCORE repository [12], which makes a data-driven photometric model possible.

Satellite range scheduling is an oversubscribed combinatorial problem [3], usually approached with multi-objective optimisation, metaheuristics or reinforcement learning [4]; for optical ground stations, a recent proposal is a genetic algorithm with tabu memory [13]. Earth observation scheduling is a large adjacent literature [5], in which cloud cover appears as an uncertainty hedged during imaging tasking [14] and recent

planners couple cloud nowcasts with the schedule [15], although always on the *operator* side. Where the structure is single-machine interval selection, exact polynomial algorithms exist: weighted interval scheduling is a textbook dynamic program [16, 17].

Professional observatories schedule around weather with queue schedulers that re-plan as conditions change [18], but assume an institutional software stack and on-site sky monitoring. Forecast verification and forecast value are mature fields [19], and statistical post-processing is essential before a forecast drives a decision [20]. Forecast value depends on the user's decision problem and not only on skill [21, 22]. Ground-based sky imaging estimates cloud from colour: the classical rule thresholds the red-to-blue ratio [23], and convolutional networks now segment cloud in whole-sky images [24], building on encoder-decoder architectures such as U-Net [25]. To our knowledge neither forecast post-processing nor sky imaging has been applied to observer-side pass planning.

3 System Design

SkyPass is a five-stage pipeline (Fig. 1). Each stage is a pure function of the preceding one, so any stage may be ablated independently, which is the mechanism by which the weather claim is evaluated.

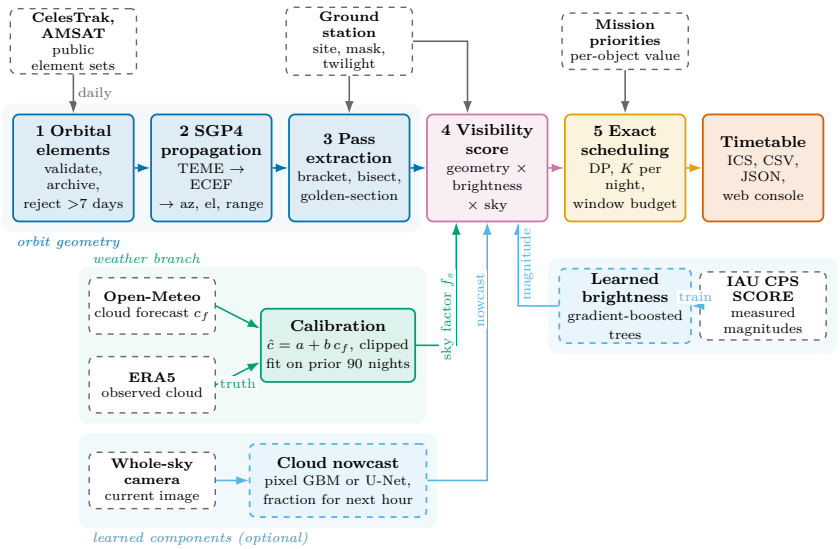


Fig. 1 SkyPass architecture. Orbit geometry (blue) turns element sets into passes; the visibility score (purple) combines geometry, brightness and sky; the weather branch (green) calibrates the cloud forecast against ERA5 reanalysis on the 90 nights before planning; the exact scheduler (amber) selects passes under the nightly limit and window budget; the timetable (vermilion) is exported as calendar, tabular and web output. Dashed grey boxes are external inputs; the dashed light-blue lane holds the optional learned components.

3.1 Element sets, propagation and pass extraction

Element sets are fetched from CelesTrak [26], checksum-validated, deduplicated per NORAD identifier keeping the newest epoch, and *archived under the date of retrieval*, since CelesTrak serves only current elements and without a local archive a published prediction cannot be repeated. Sets older than $\tau_{\max}$ are rejected. Positions come from the reference SGP4 implementation [8]; the TEME vector is rotated to ECEF by Greenwich mean sidereal time and converted to topocentric azimuth, elevation and range on the WGS-84 ellipsoid. Polar motion is neglected, being far under the SGP4 error budget, and refraction is reported but not applied to the horizon mask, so results do not depend on assumed surface conditions.

Because cost is dominated by propagator calls, pass extraction minimises them (Fig. 2). Elevation is sampled on a coarse grid to *bracket* horizon crossings; since elevation within a pass is smooth and unimodal, a sign change of $\varepsilon(t) - \varepsilon_{\text{mask}}$ between adjacent samples brackets exactly one crossing. Brackets are refined by bisection to 0.1 s and culmination by golden-section search. The coarse step is $P/240$ bounded to 5 s to 30 s, so short-period orbits, whose passes are shortest, are sampled most finely.

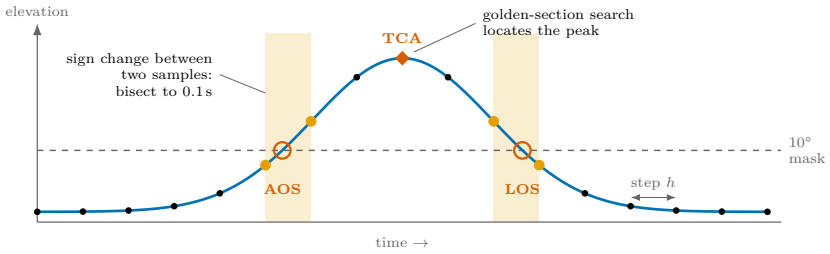


Fig. 2 Pass extraction (illustrative curve). Elevation is sampled every h seconds (black). A sign change of elevation minus the 10° mask between neighbouring samples (amber) brackets a rise or set, which bisection refines to 0.1 s (AOS, LOS); golden-section search locates culmination (TCA).

3.2 Visibility scoring

Each pass receives a score $S \in [0, 1]$ combining a geometric quality term with multiplicative gates:

$$S = \underbrace{(w_e f_e + w_d f_d)}_{\text{geometry}} \cdot f_m \cdot f_s, \tag{1}$$

where $f_e = \min(\varepsilon_{\max}/60°, 1)$ and $f_d = \min(T/480\,\text{s}, 1)$ saturate culmination elevation and duration, f_m is a photometric factor and f_s a sky-condition factor.

The multiplicative form is deliberate: an eclipsed pass or one in a bright sky scores zero irrespective of its geometry, and expected cloud scales the score down in proportion, which an additive score cannot represent. For optical work the object has to be sunlit and the observer in darkness; Earth shadow uses the dual-cone construction [1], and apparent magnitude follows the standard diffuse-sphere relation [10]

$$m = m_{\text{std}} + 5 \log_{10} \frac{d}{1000\,\text{km}} - 2.5 \log_{10} F(\psi), \tag{2}$$

with $F(\psi) \propto (\pi - \psi)\cos\psi + \sin\psi$ normalised to $F(0) = 1$ (Fig. 3). In radio mode the illumination and photometric terms are dropped.

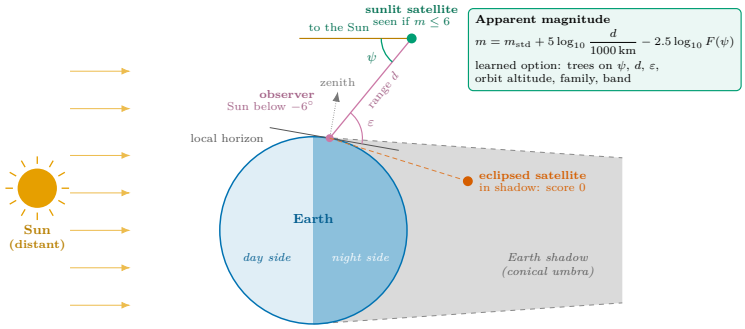


Fig. 3 Observation geometry (schematic, not to scale). A pass is observable when the satellite is sunlit, outside the conical Earth shadow, while the Sun is more than 6° below the observer's horizon. Its apparent magnitude then depends on the range d and the phase angle ψ between the directions to the Sun and to the observer (2); an eclipsed satellite scores zero whatever its elevation ε.

3.3 Weather integration and calibration

The sky factor is $f_s = 1 - \hat{c}$, where $\hat{c}$ is the expected cloud fraction at culmination, interpolated linearly between hourly samples. It is *not* the raw forecast, which is heavily over-confident as a predictor. We fit

$$\hat{c} = \mathbb{E}[c \mid c_f] = a + b\,c_f \tag{3}$$

by least squares on a window that *ends before* the period being planned, restricted to local night hours, and $\hat{c}$ is clipped to $[0, 1]$. The slope b quantifies the signal: $b \approx 1$ means the forecast can be used directly, $b \approx 0$ that the calibrated value falls back to site climatology. For the threshold value model the planner instead weights each pass by the calibrated probability of clear sky, $P(c \le \theta \mid c_f)$, estimated in ten forecast bins on the same window and shrunk toward the site's base rate with 20 pseudo-observations, so each value model is paired with the planner that maximises its expectation.

3.4 Conflict resolution

A single-aperture station observes one object at a time and needs a setup gap g between targets, so choosing the best mutually compatible subset of passes is weighted interval scheduling. Writing $v_j = S_j\pi_j$ for the value of pass j (score times priority), indexing passes $1, \ldots, n$ by finish time and letting $p(j)$ be the largest $i < j$ compatible with j (0 if none), the textbook recurrence [16, 17] gives the exact optimum in $O(n \log n)$.

That alone is insufficient: unconstrained, a station observes every candidate and weather-awareness has nothing to select. Real stations are limited by operator and

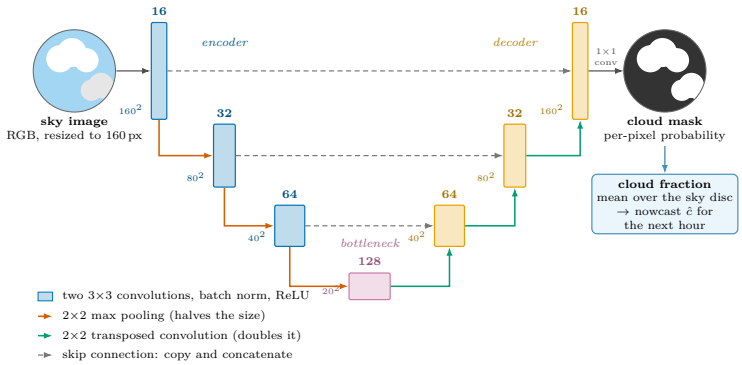


Fig. 4 The cloud-segmentation U-Net. The encoder halves the image three times while doubling the channels, the decoder restores the resolution, and skip connections carry fine detail across. A 1×1 convolution gives the per-pixel cloud probability, whose mean over the sky disc is the nowcast.

instrument time, so a capacity budget of K observations per night adds a cardinality dimension:

$$\begin{aligned}\mathrm{OPT}[j][k] = \max\big(&\mathrm{OPT}[j-1][k],\\ &v_j + \mathrm{OPT}[p(j)][k-1]\big),\end{aligned} \tag{4}$$

with $\mathrm{OPT}[0][k] = \mathrm{OPT}[j][0] = 0$. This is exact. The table has one entry for each of the n candidate passes and each count up to the nightly limit K, and each entry is filled in constant time, so after the sort the cost is $O(nK)$: doubling either the candidates or the limit doubles the work. Observing nights are disjoint, so applying the budget per night and solving each independently stays exact. Nights are defined in local solar time, so an evening and the small hours that follow form one session.

3.5 Learned components

Two optional models replace hand-built approximations (dashed in Fig. 1); the planner runs unchanged without them.

The cloud forecast is hourly and, as Section 5.4 shows, cannot discriminate within a night; a whole-sky camera can. Two segmenters label each pixel of a camera image as cloud or sky. The first is a U-Net [25] with three down-sampling stages and 16 base channels (Fig. 4), trained with a binary cross-entropy masked so that pixels outside the fisheye contribute nothing, and with rotations by multiples of 90° and flips, because the sky has no preferred azimuth. The second is a gradient-boosted classifier [27] on fifteen per-pixel features: the colour channels, the red-to-blue ratio, the normalised blue-red difference, saturation and brightness, and the local mean and spread of two of them at two scales. The cloud fraction of the image replaces the calibrated forecast for the next hour; being a measurement, it is not itself calibrated.

The magnitude gate of (2), with a tabulated standard magnitude, is an ordering rather than a prediction. The learned alternative, gradient-boosted regression trees,

takes the phase angle, range, elevation, solar elevation and orbital altitude at culmination, the sphere prediction, the satellite family recognised from the catalogue name and the observing band; the family is known for every catalogued object, so the model applies to satellites never measured.

4 Experimental Setup

The catalogue comprises 635 unique objects from CelesTrak GP groups, checksum-validated and deduplicated, observed from seven stations spanning distinct cloud climatologies: Chennai and Bengaluru (monsoon), Leh (high-altitude arid), Tucson (desert), Singapore (equatorial), and Reykjavík and Svalbard (maritime polar). Unless stated otherwise, the mask is 10°, the dark-sky threshold −6° and the setup gap 5 min. Timing is checked against a brute-force 1 s dense scan and against Skyfield [28], which shares only the reference SGP4 core with SkyPass. All runs are single-threaded under Python 3.13.5 with `sgp4` 2.27.

All three weather products come from Open-Meteo [29]: the operational forecast, archived past forecasts at lead times of 1–7 days, and ERA5 reanalysis [6] as verification truth. One methodological point silently invalidates the evaluation if overlooked: the archive endpoint has to be pinned to the ERA5 model, since at its default it returns a best-match series which, for recent dates, *is* the archived forecast, so verifying against it compares a forecast with a copy of itself and reports a spurious skill of 1.0.

Measured magnitudes come from SCORE [12], whose rows carry the SatChecker geometry of each observation. Removing rows flagged as discrepant or unilluminated, those below the horizon, and the solar sail ACS 3, which alone supplies most of the repository, leaves 17 216 observations of 2709 satellites in 13 families, cross-validated in five folds grouped by satellite and split in time at 2025-07. Sky images are the 400 labelled whole-sky images of WSISEG [24], from one camera, cross-validated in five contiguous blocks because consecutive frames are near-duplicates. Every method sees identical training data. The camera baselines are the red-to-blue ratio rule [23] at its classical threshold of 0.6 and with the threshold tuned on the training images; the photometric baselines are the sphere model with its tabulated standard magnitudes and with one standard magnitude per family fitted on the training data.

A generative AI assistant (Claude, Anthropic) was used to help edit and restructure the manuscript text and to write the scripts that build it; the authors reviewed all such output and take responsibility for the content.

5 Results

5.1 Pass-prediction accuracy and cost

Fig. 5a compares SkyPass against both references over 635 objects and 1761 matched passes. Against Skyfield, mean agreement is 0.060 s on acquisition and 0.055 s on culmination, and culmination elevation agrees to 0.0034°. The apparent 0.504 s “deviation” from the dense scan is not SkyPass error: a 1 s grid locates a crossing only to within its own step, so it lags the true crossing by a mean of 0.5 s, and the refined

estimator is therefore more accurate than the brute-force reference it is conventionally compared against. The cost difference is large: 2 311 148 propagator evaluations against 54 864 000, a factor of 23.7. Detection recall is 99.78%, the residual being short grazing transits that peak barely above the mask.

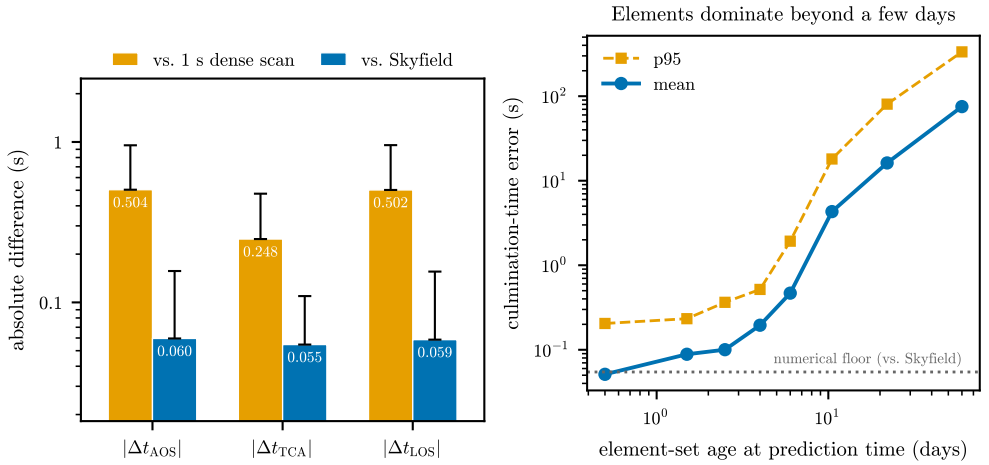


Fig. 5 (a) Mean timing difference at acquisition, culmination and loss of signal, against the 1 s dense scan and against Skyfield; whiskers mark the 95th percentile. (b) Culmination-time error against element-set age (mean and 95th percentile, logarithmic axes); the dotted line is the numerical agreement with Skyfield.

5.2 Element-set limits on prediction accuracy

Sub-second numerical agreement is meaningful only if the elements support that precision. The median element-set age in the operational catalogue was 0.42 days (90.2% under one day). CelesTrak and AMSAT sets differ by a mean of 0.041 s in culmination over 308 matched passes, but by a median of 0.00 s: the providers largely redistribute the same upstream sets, so this is not an independent check. Re-predicting the same passes from progressively older sets of 60 low-Earth objects gives 33 223 aged comparisons (Fig. 5b). Within a day of epoch the elements contribute 0.05 s, the same order as the 0.055 s numerical agreement of Section 5.1, so neither term dominates; degradation is then sharply super-linear and heavy-tailed, the mean reaching 4.3 s (median 0.70 s) at 7–14 days and 75.3 s (median 13.69 s) beyond a month, so a 5–7 day staleness cutoff suits timing-critical work and daily element refresh is the largest single lever on prediction accuracy.

5.3 Skill of the cloud forecast

Weather-aware planning is justified only if the forecast is informative at the lead times over which an observer plans. Archived forecasts were verified against ERA5 at all seven stations over 60 days, scoring the operational decision of whether an hour is clear enough to observe at a 30% cloud threshold. Because accuracy misleads when

clear hours are rare, we report the Heidke skill score, a chance-corrected measure in wide use for categorical verification [30] (Fig. 6). Skill decays from HSS 0.347 at one day to 0.289 at three and 0.114 at seven, beating persistence (0.206) up to three days ahead but not from four days on, so a weather-aware planner is best run on a rolling short horizon. Skill is also strongly site-dependent, the spread between stations being comparable to the decay with lead time and lowest at the two monsoon stations: the expected benefit is a local property rather than a universal constant.

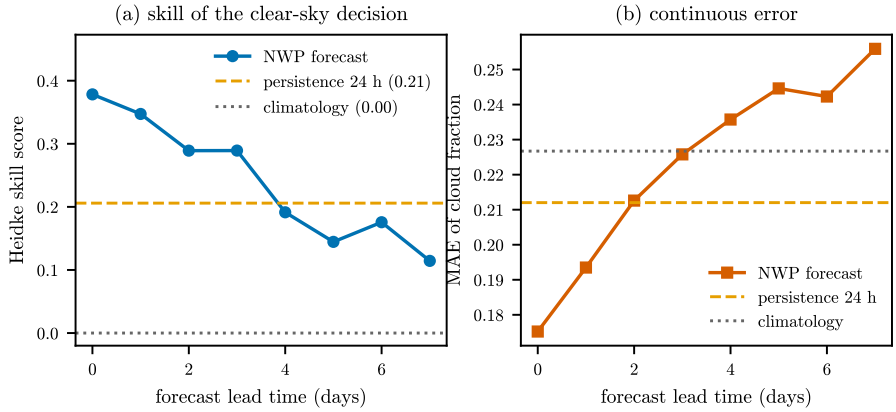


Fig. 6 Skill of the archived cloud forecast against ERA5, pooled over seven stations. (a) Heidke skill score of the clear-hour decision by lead time, with persistence and climatology baselines. (b) Mean absolute error of cloud fraction.

5.4 Value of weather-aware scheduling

Five planners select from an identical candidate set, drawn from 300 objects, at each of six stations over 60 days (Svalbard has no observable passes), and every timetable is scored against ERA5. Since the valuation of a clouded observation is a modelling choice, we report both an *expected value* model, $\sum_j v_j(1-c_j)$, and a *threshold* model, $\sum_j v_j \mathbf{1}[c_j \le \theta]$ with $\theta = 30\%$. Two operating rules are compared (Fig. 7): a fixed nightly quota, and a window budget that lets the station skip nights.

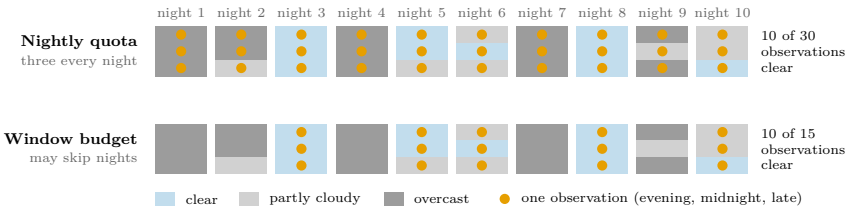


Fig. 7 The two operating rules, illustrated (not data). Each column is one night split into evening, midnight and late; cloud tends to persist through a night. Under a nightly quota the forecast can only reorder passes within a night; under a window budget it decides which nights to use.

Under a fixed nightly quota the forecast yields no benefit: −1.5% and −0.2%

under the two models, against oracle headroom of 12.0% and 31.6%, and the station-bootstrap intervals of both gains include zero (−2.7 to 2.0% and −11.6 to 3.5%). The same holds per station: the oracle shows substantial headroom at every station, while the forecast-driven planner gains at most a few per cent and loses at three, and no forecast lead time from one to seven days changes this (Fig. 8a). The forecast is not uninformative: forecast and observed cloud correlate at $r = 0.41$ at the candidate-pass timestamps, and the clearest-forecast decile does find clearer skies, reducing observed cloud by 0.209. The limitation is that a quota permits only *reordering within a night*, which two measured properties defeat: cloud is autocorrelated over the hours a session spans, the between-night standard deviation being 2.2× the within-night value, while the ninetieth-percentile pass carries 1.8× the mean weather-free value. The score being a product, selecting for clear sky surrenders base value, an unfavourable exchange when pass quality is the more variable factor.

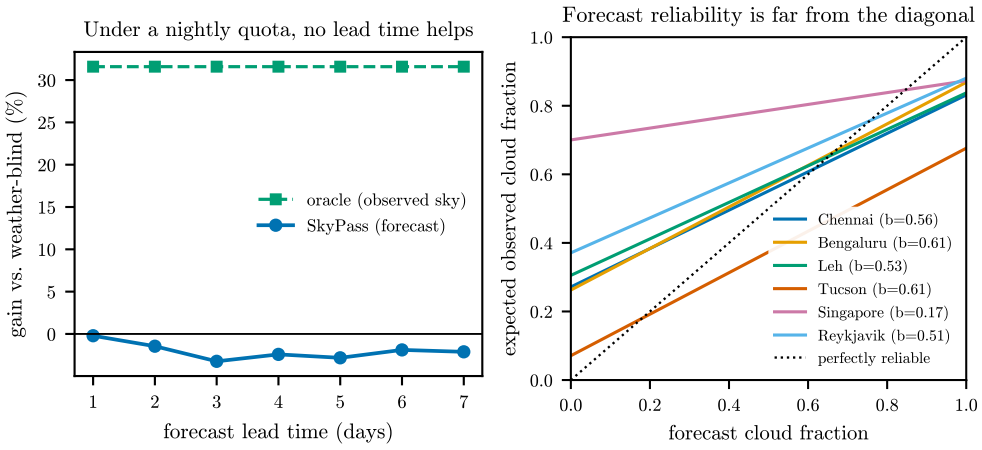


Fig. 8 (a) Gain under a nightly quota against forecast lead time, for the forecast-driven planner and the oracle. (b) Fitted forecast reliability per station; a perfectly reliable forecast would lie on the dotted diagonal.

Given the same cap plus a window budget, so the station may *skip* a night, the result reverses (Fig. 9): which nights to use is a day-scale decision, the scale at which the forecast retains skill. We report the central of the three budget levels as the headline; Fig. 9 shows all four regimes. Observing on 50% of nights (90 observations), realised yield improves by 19.0% and 37.6% (95% station-bootstrap intervals 9.2–24.9% and 3.1–80.4%), recovering 41% and 35% of the attainable oracle gains; clear-sky passes rise from 13.9% to 23.1%, and a quarter-of-nights budget raises the threshold-model gain to 55.0% (24.7–144.5%).

Calibration raises the gain in every regime. Under the quota, supplying the model cloud field straight to the score costs −9.2% relative to disregarding weather; under the half budget the raw field still helps (8.0% and 29.9%) but less than the calibrated one. Fitted slopes (Fig. 8b) lie between 0.17 and 0.61 and intercepts between 0.07 and 0.70, so clear forecasts are followed by more cloud than predicted and cloudy ones

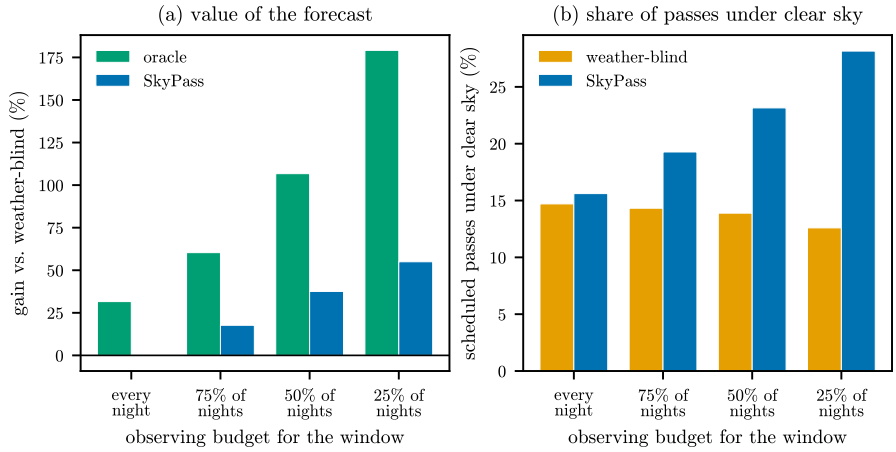


Fig. 9 Weather-awareness is worth nothing to a station obliged to observe every night, and a great deal to one that may choose its nights. (a) Gain over the weather-blind exact planner under the same budget, threshold value model; the oracle plans with the observed cloud and bounds what any forecast could achieve. (b) The share of scheduled passes that fell under clear sky.

by less. Calibration (3) shrinks the forecast toward climatology in proportion to its demonstrated reliability.

Repeating the evaluation with epoch-correct geometry, that is, with the element set current on each night, raises the threshold-model gain under the quota from −0.2% to 9.2% and at a quarter budget from 55.0% to 75.9%, but lowers it at the half budget from 37.6% to 33.2%, the expected-value gain there being unchanged at 19.0%. Back-propagated geometry therefore does not inflate the result systematically, although it is not uniformly conservative.

5.5 Conflict resolution in practice

On 2000 randomised instances small enough to enumerate exhaustively, the dynamic program matched the optimum in 100.00% of cases. Greedy selection *by the score itself* attains 99.95% on average, so the dynamic program buys little in the average case; its advantage is being exact at negligible cost with no adverse case (Fig. 10a). The rules an operator applies fare worse, optimising a proxy rather than the objective: selection by highest culmination attains 95.39% on average, is optimal in only 62.7% of instances and falls to 9.4% at worst. An order-based genetic algorithm of the type common in the satellite-tasking literature (population 60, 120 generations) reaches 99.67% of the optimum but needs roughly 38 922× more time, a ratio that depends on that search budget; once the observer-side problem is recognised as interval scheduling, an exact polynomial solution is available and a metaheuristic is unnecessary. Scheduling is not the bottleneck in any case, the dynamic program handling 100 000 candidates in 285.3 ms against the seconds to minutes propagation needs (Fig. 10b).

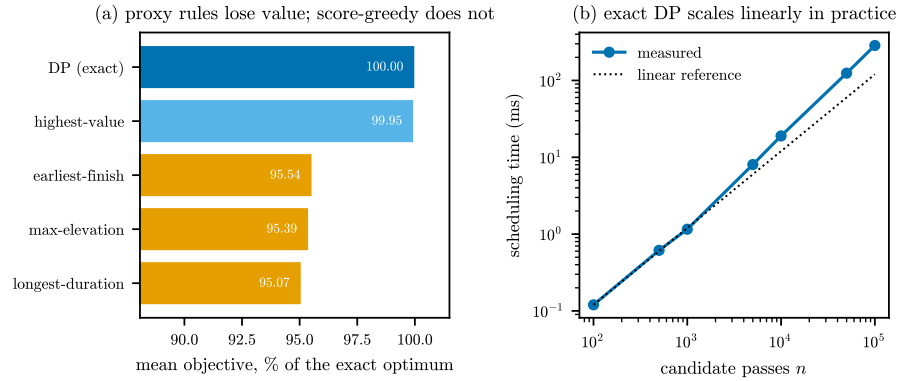


Fig. 10 (a) Mean objective of greedy selection rules as a percentage of the exact optimum over 2000 random instances. (b) Run time of the exact dynamic program against the number of candidate passes, with a linear reference.

5.6 End-to-end behaviour

The funnel is steep and physically interpretable (Fig. 11): at the demonstration site 12 449 geometric passes reduce to 8093 sunlit, 1297 in darkness, 367 bright enough to detect, 87 above the score floor once forecast cloud is weighted in, and 26 scheduled. Svalbard (78.2° N) returns the most geometric passes of any station yet *zero* observable ones, the Sun never falling below −6° there in the planned week.

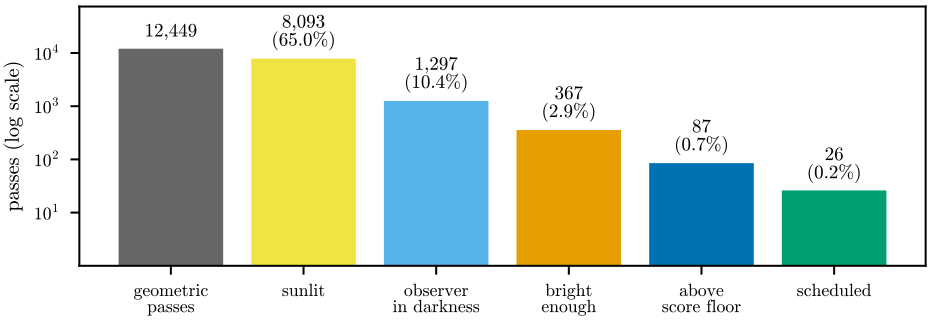


Fig. 11 Visibility funnel for the demonstration station: of 12 449 geometric passes over 7 days, 26 survive to the timetable. Cloud enters the score as a continuous weight rather than a hard cut, so the penultimate stage is the score floor.

Planning 635 objects over 7 days takes 109.4 s single-threaded, dominated by propagation. Varying w_e from 0.3 to 0.7 leaves a Jaccard overlap of at least 0.89 with the default timetable, falling to 0.70 once elevation dominates ($w_e \geq 0.9$).

5.7 Cloud nowcasting from whole-sky images

The classical colour-ratio rule at its fixed threshold of 0.6 does not transfer to this camera: it labels 72.6% of pixels correctly and misestimates the image cloud fraction by 27.0 percentage points on average, since the ratio depends on the colour response

of the instrument. Tuning the threshold on the training images (0.74 in every fold) recovers most of the gap, to 92.3% of pixels and 3.5 points. Both learned segmenters are better again (Fig. 12). The U-Net reaches a cloud intersection-over-union of 0.927 and the pixel classifier 0.945, and both estimate the cloud fraction to within 1.2 points. The pixel classifier was the more accurate segmenter in every fold, plausibly because its texture features resolve cloud edges that the U-Net, trained at 160 pixels, smooths; for the cloud fraction the two are equivalent, and SkyPass deploys the pixel classifier, which needs no deep-learning framework. All images come from one camera.

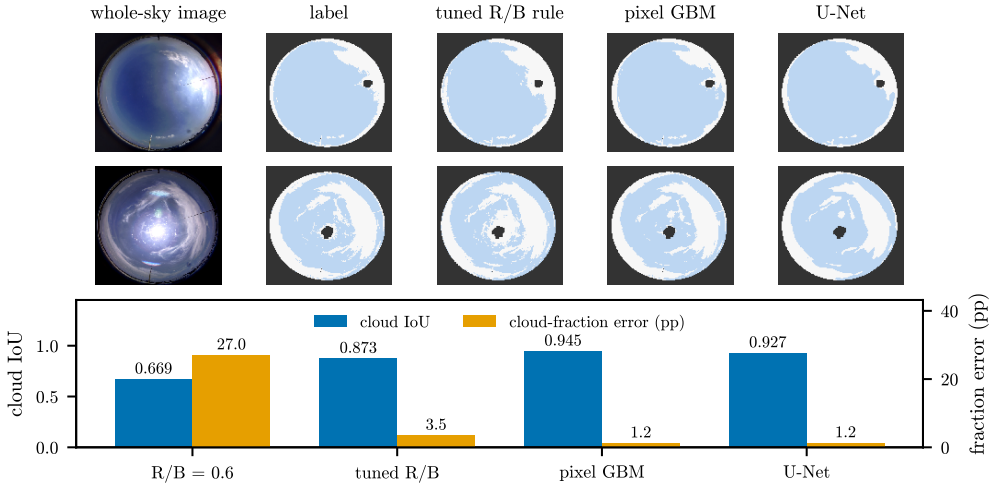


Fig. 12 Cloud segmentation of whole-sky images. Top: two held-out WSISEG images with their labels and the predictions of the tuned colour-ratio rule, the pixel classifier and the U-Net (cloud white, sky blue, undefined pixels dark). Bottom: cloud intersection-over-union and the error of the image cloud fraction for each method under blocked five-fold cross-validation.

5.8 Learned satellite brightness

For satellites absent from training, the learned model predicts apparent magnitude with a mean absolute error of 0.83 mag, against 1.51 for the sphere with one fitted standard magnitude per family and 1.86 for the sphere with tabulated values (Fig. 13a), and as a visibility decision at the 6 mag limit its balanced accuracy is 0.80 against 0.66 and 0.59. Orbital altitude and family carry most of the skill (Fig. 13b), standing in for size, attitude and mitigation design, which a sphere with one albedo cannot represent.

The advantage does not survive a change of period. Trained before 2025-07 and tested after, the learned model's error (2.00 mag) equals that of the fitted sphere (1.99) and its balanced visibility accuracy falls to 0.51. It stays better on the 6 families absent from training (2.69 against 3.29 mag) but is worse on established ones (1.60 against 1.26); in Starlink, the largest family, the errors are 2.76 and 1.57 mag, consistent with design changes between generations. The model interpolates across satellites but not

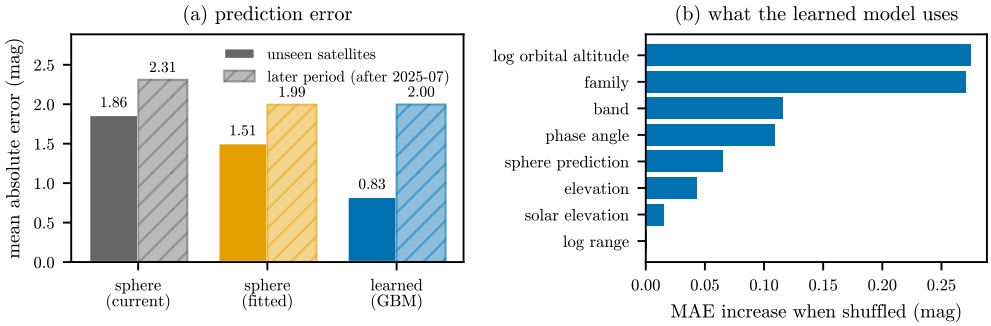


Fig. 13 Learned satellite brightness on 17 216 SCORE observations of 2709 satellites. (a) Mean absolute magnitude error for satellites absent from training (grouped cross-validation, dark bars) and for observations after 2025-07 with training on earlier ones (light bars). (b) Permutation importance in the learned model: the increase in error when one input is shuffled.

across time, so the sphere stays the default and the learned model is an option to be retrained as SCORE grows.

6 Discussion and Limitations

Cloud cover is an imperfect proxy for observing conditions: total cloud fraction on a reanalysis grid cell of order 25 km describes one line of sight only coarsely, and transparency, seeing, cloud height and aerosol are not modelled. A camera nowcast can be read from an image with small error (Section 5.7), the route to the one regime in which the forecast gives no benefit; its planning value could not be measured, because public camera archives do not cover the evaluation nights, and its transfer to other cameras is untested.

The weather-value gains pool six stations and 60 nights in one season; with so few clusters the station-bootstrap intervals are approximate and probably too narrow, and the calibrated-minus-raw difference was not tested separately.

Neither value model captures partial credit; the scheduler accepts a station's own value function unchanged. One instrument observing one object at a time is assumed, so multiple apertures or a station network would need a different algorithm. The learned photometry drifts as designs change (Section 5.8); the ageing measurement is single-season and limited to low Earth orbit.

7 Conclusion

SkyPass integrates pass prediction, visibility analysis, cloud-forecast reasoning and exact conflict resolution into one planner for a small ground station. A forecast is informative about which *nights* merit observation and much less so within a night, so a planner under a fixed nightly quota gains nothing, whereas one with a budget to allocate gains 19.0% (expected value) and 37.6% (clear-sky count); calibration against reanalysis raises the gain in every regime tested. Forecast value is a property of the

decision it informs, a finding that should carry over to other small facilities, though only one setting is tested here.

The learned components follow the same principle: segmentation of a whole-sky image supplies the sub-hourly signal a quota-bound station needs, and learned photometry predicts magnitudes for unseen satellites far better than the sphere, but only while its training data describe the satellites in orbit.

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Statements and Declarations

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No funding was received for conducting this study. The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval and consent

Not applicable: the study involves no human participants, human data or animals.

Data and code availability

SkyPass is open source under the MIT licence at https://github.com/rohithprem18/skypass, with the element sets used and the raw experiment outputs behind every reported value. Weather series are retrieved from Open-Meteo's public archive by the released scripts; historical element sets for the ageing and epoch-correct experiments come from Space-Track, which requires a free account, and are not redistributed. The SCORE magnitudes (CC-BY 4.0) and the WSISEG images are downloaded by the experiment scripts from their providers and are not redistributed; the trained models are rebuilt from them by the same scripts.

References

[1] Vallado, D.A.: Fundamentals of Astrodynamics and Applications, 5th edn. Microcosm Press, Hawthorne, CA (2022)

[2] Acciarini, G., Baydin, A.G., Izzo, D.: Closing the gap between SGP4 and high-precision propagation via differentiable programming. Acta Astronautica **226**, 694–701 (2025) https://doi.org/10.1016/j.actaastro.2024.10.063

[3] Li, S., Yu, Q., Ding, H.: Reviews and prospects in satellite range scheduling problem. Autonomous Intelligent Systems **3**(1), 9 (2023) https://doi.org/10.1007/s43684-023-00054-6

[4] Ou, J., *et al.*: Deep reinforcement learning method for satellite range scheduling problem. Swarm and Evolutionary Computation **77**, 101233 (2023) https://doi.org/10.1016/j.swevo.2023.101233

[5] Wang, X., Wu, G., Xing, L., Pedrycz, W.: Agile Earth observation satellite scheduling over 20 years: Formulations, methods, and future directions. IEEE Systems Journal **15**(3), 3881–3892 (2021) https://doi.org/10.1109/jsyst.2020.2997050

[6] Hersbach, H., *et al.*: The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society **146**(730), 1999–2049 (2020) https://doi.org/10.1002/qj.3803

[7] Hoots, F.R., Roehrich, R.L.: Models for propagation of NORAD element sets. Technical Report Spacetrack Report No. 3, Aerospace Defense Command, United States Air Force (1980)

[8] Vallado, D.A., Crawford, P., Hujsak, R., Kelso, T.S.: Revisiting spacetrack report #3. In: AIAA/AAS Astrodynamics Specialist Conference and Exhibit, pp. 2006–6753 (2006). https://doi.org/10.2514/6.2006-6753

[9] Kovář, P., Puričer, P., Kovářová, K.: Study of the two-line element accuracy by 1U CubeSat with a GPS receiver. Sensors **22**(8), 2902 (2022) https://doi.org/10.3390/s22082902

[10] Fankhauser, F., Tyson, J.A., Askari, J.: Satellite optical brightness. The Astronomical Journal **166**(2), 59 (2023) https://doi.org/10.3847/1538-3881/ace047

[11] Hainaut, O.R.: Large or bright satellite constellations. Astronomy & Astrophysics **712**, 241 (2026) https://doi.org/10.1051/0004-6361/202660006

[12] IAU Centre for the Protection of the Dark and Quiet Sky: Satellite Constellation Observation Repository (SCORE). https://score.cps.iau.org/. Data released under CC-BY 4.0; accessed 2026-10-21 (2026)

[13] Zhai, S., Cheng, W., Zhang, T.: A resource scheduling algorithm for optical ground stations supporting satellite observation tasks. Aerospace **13**(10), 892 (2026) https://doi.org/10.3390/aerospace13100892

[14] Wang, X., Song, G., Leus, R., Han, C.: Robust Earth observation satellite scheduling with uncertainty of cloud coverage. IEEE Transactions on Aerospace and Electronic Systems **56**(3), 2450–2461 (2020) https://doi.org/10.1109/taes.2019.2947978

[15] Wang, X., Jin, Z., Wang, C., A, Z.: Prediction-enhanced cloud-aware scheduling for micro agile maritime observation satellite. Journal of Aerospace Information Systems **23**(9), 847–863 (2026) https://doi.org/10.2514/1.I011843

[16] Kleinberg, J., Tardos, É.: Algorithm Design. Pearson/Addison-Wesley, Boston (2006)

[17] Cormen, T.H., Leiserson, C.E., Rivest, R.L., Stein, C.: Introduction to Algorithms, 4th edn. MIT Press, Cambridge, MA (2022)

[18] Xie, H., Kang, Z., Jiang, X.: AstroSA: An astronomical observation scheduler assessment framework in Python. Astronomy and Computing **47**, 100806 (2024) https://doi.org/10.1016/j.ascom.2024.100806

[19] Brown, B., *et al.*: The model evaluation tools (MET): More than a decade of community-supported forecast verification. Bulletin of the American Meteorological Society **102**(4), 782–807 (2021) https://doi.org/10.1175/bams-d-19-0093.1

[20] Vannitsem, S., *et al.*: Statistical postprocessing for weather forecasts: Review, challenges, and avenues in a big data world. Bulletin of the American Meteorological Society **102**(3), 681–699 (2021) https://doi.org/10.1175/bams-d-19-0308.1

[21] Richardson, D.S.: Skill and relative economic value of the ECMWF ensemble prediction system. Quarterly Journal of the Royal Meteorological Society **126**(563), 649–667 (2000) https://doi.org/10.1002/qj.49712656313

[22] Raeth, K., Ludwig, N.: Forecast skill is not decision skill: Evidence from weather-dependent decision tasks. Meteorological Applications **33**(5), 70260 (2026) https://doi.org/10.1002/met.70260

[23] Long, C.N., Sabburg, J.M., Calbó, J., Pagès, D.: Retrieving cloud characteristics from ground-based daytime color all-sky images. Journal of Atmospheric and Oceanic Technology **23**(5), 633–652 (2006) https://doi.org/10.1175/JTECH1875.1

[24] Xie, W., Liu, D., Yang, M., Chen, S., Wang, B., Wang, Z., Xia, Y., Liu, Y., Wang, Y., Zhang, C.: SegCloud: a novel cloud image segmentation model using a deep convolutional neural network for ground-based all-sky-view camera observation. Atmospheric Measurement Techniques **13**(4), 1953–1961 (2020) https://doi.org/10.5194/amt-13-1953-2020

[25] Ronneberger, O., Fischer, P., Brox, T.: U-Net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention (MICCAI 2015). Lecture Notes in Computer Science, vol. 9351, pp. 234–241. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-24574-4_28

[26] Kelso, T.S.: CelesTrak: Current NORAD Two-Line Element Sets. https://celestrak.org/. Accessed 2026-08-29 (2026)

[27] Friedman, J.H.: Greedy function approximation: A gradient boosting machine. The Annals of Statistics **29**(5), 1189–1232 (2001) https://doi.org/10.1214/aos/1013203451

[28] Rhodes, B.: Skyfield: High Precision Research-Grade Positions for Planets and Earth Satellites. Python package, version 1.54, https://rhodesmill.org/skyfield/ (2026)

[29] Zippenfenig, P.: Open-Meteo.com Weather API. https://open-meteo.com/ (2024). https://doi.org/10.5281/zenodo.7970649

[30] Ebert, P.A., Milne, P.: Methodological and conceptual challenges in rare and severe event forecast verification. Natural Hazards and Earth System Sciences **22**(2), 539–557 (2022) https://doi.org/10.5194/nhess-22-539-2022